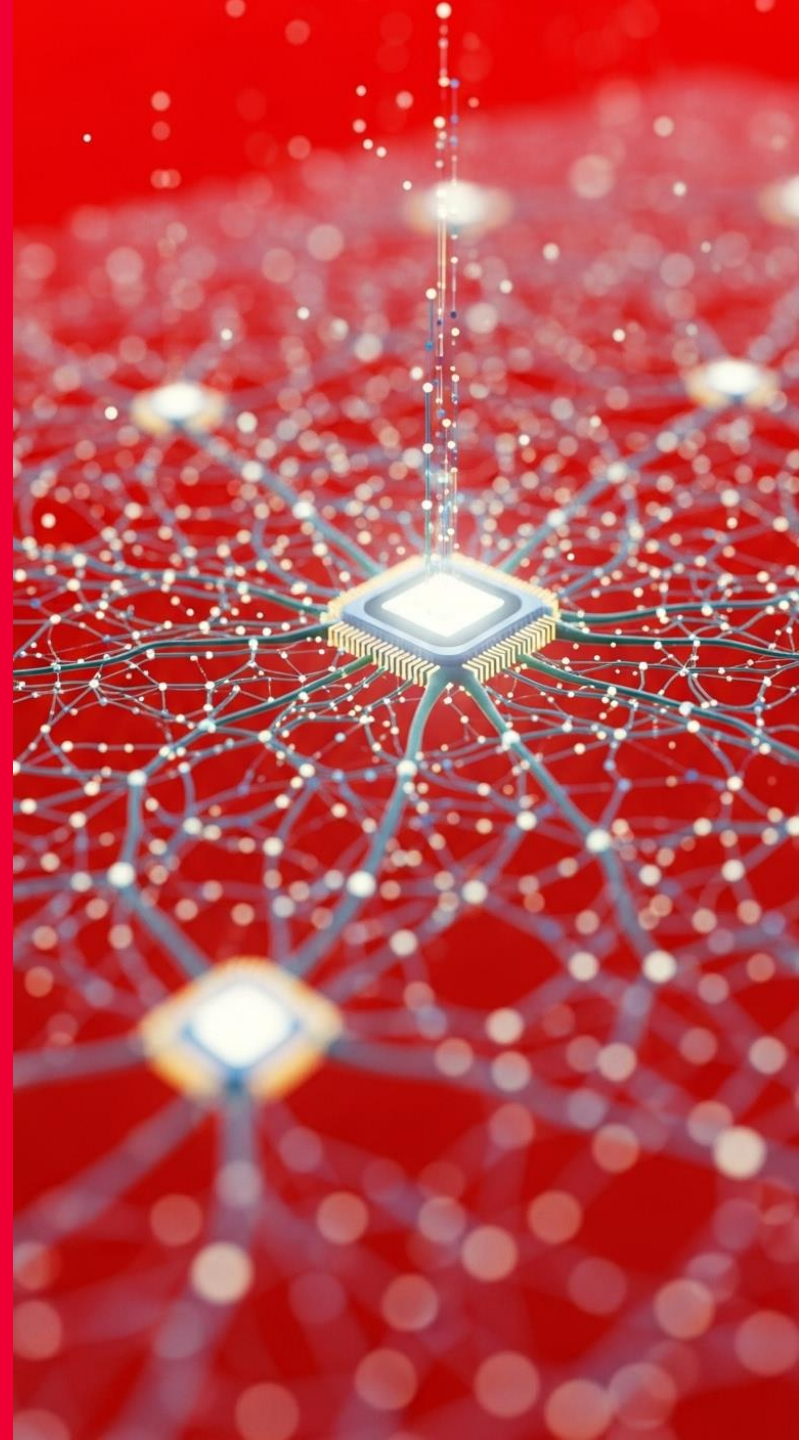


Cambridge Summer School

Artificial Intelligence fundamentals

Dr Mariano Felice

5th August 2026



Outline

What is AI?

- Definition

- Brief history

Types of AI

- Symbolic methods

- Data-driven methods

Machine Learning

- Definition

- Typical pipeline

- Supervised vs. unsupervised learning

Supervised Learning

- Classification

- Regression

Evaluation

- Evaluation metrics

- Overfitting and underfitting

The origins of AI (1950–1960)

Alan Turing: The Father of AI



Alan Turing
1912 - 1954

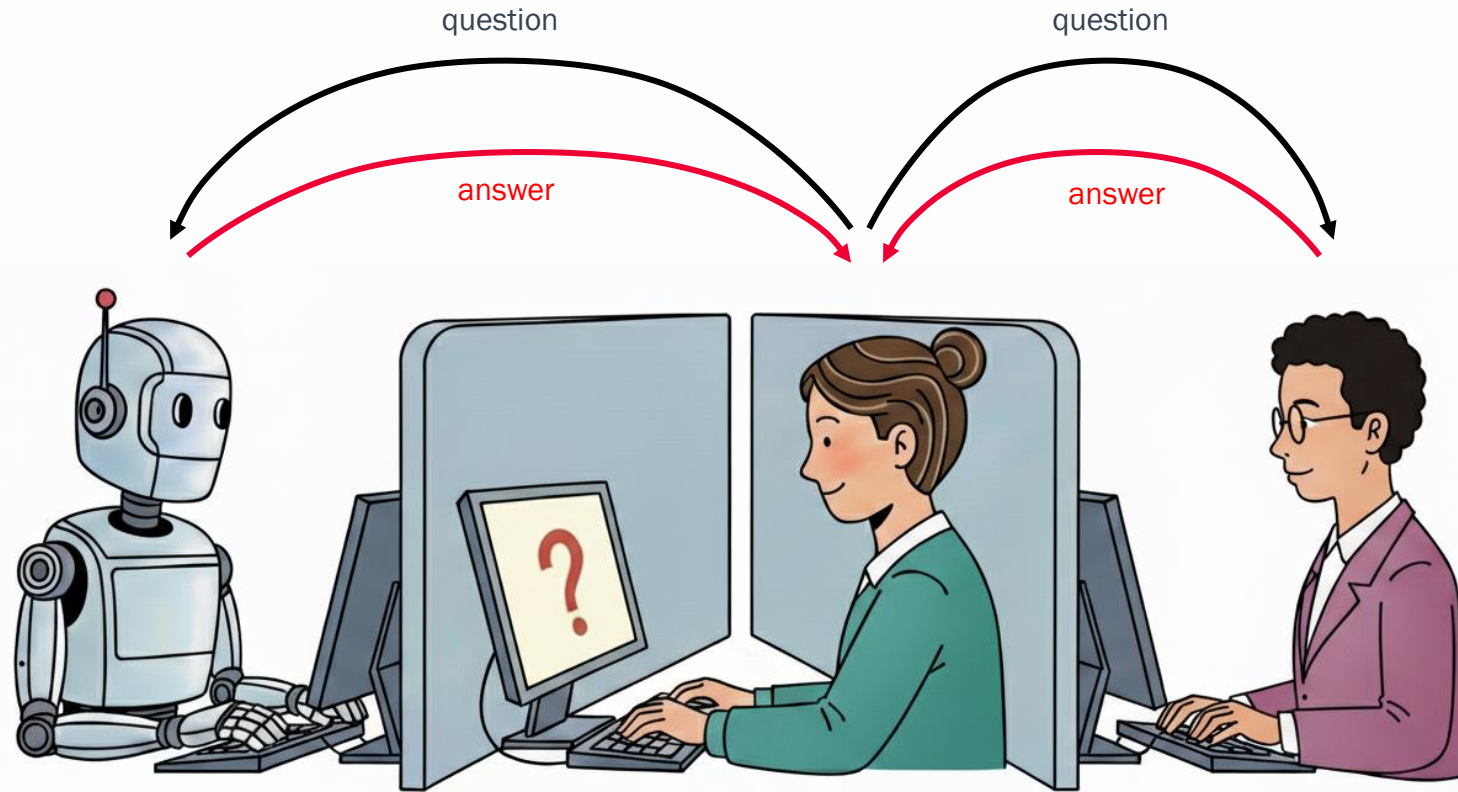
mathematician, computer pioneer
and code breaker

Turing's seminal paper "Computing Machinery and Intelligence" (1950).

Intelligence is measured through the ability to talk like a human.

If a machine can perform every intellectual task a human can, we have no logical right to deny it "intelligence".

The Imitation Game (a.k.a. Turing Test)



A person talks with both a human and a machine. If the person can't tell which is which, the machine is seen as intelligent.

Over to you!

Which of these texts was generated by AI?

A

Society has a responsibility to take care of the environment. We all live on this planet, so it's important for us to do our part to keep it clean and healthy. There are many things we can do to be environmentally responsible. We should recycle our waste, such as paper, plastic, and cans, instead of throwing them away. It's also important to save energy by turning off lights and appliances when we're not using them. Using public transportation or carpooling can help reduce pollution from cars. We should also avoid wasting water and try to conserve it. Planting trees and taking part in community clean-up activities are great ways to contribute to a cleaner environment. By being environmentally responsible, we can make a positive impact on our planet and ensure a better future for ourselves and future generations.

B

Every country in the world suffers environmental catastrophes. This is a natural consequence of the struggle between development and environment. If a country decided to live isolated from the rest of the world, living on what it can naturally grow and produce, it surely wouldn't be highly polluted. But we all want exotic food and technological items from all over the world, so we have to pay the price. Investing on electrical transport would benefit the environment a lot. Even more if this electricity came from a natural source of energy like wind, rivers and solar boards. We also have to take care of our rivers and seas. We all have heard about factories throwing highly toxic substances to rivers, without minimizing their poisoning effects. A really strict law should be applied to fine these factories and make them change their policy.

Over to you!

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Society has a responsibility to take care of the environment. We all live on this planet, so it's important for us to do our part to keep it clean and healthy. There are many things we can do to be environmentally responsible. We should recycle our waste, such as paper, plastic, and cans, instead of throwing them away. It's also important to save energy by turning off lights and appliances when we're not using them. Using public transportation or carpooling can help reduce pollution from cars. We should also avoid wasting water and try to conserve it. Planting trees and taking part in community clean-up activities are great ways to contribute to a cleaner environment. By being environmentally responsible, we can make a positive impact on our planet and ensure a better future for ourselves and future generations.

AI

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human

Over to you!

Turing said that if a machine talks like a human, we should treat it as intelligent. Do you agree, or is there something “extra” inside us that a machine can never copy?

Does ChatGPT pass the Turing Test?

The Dartmouth Workshop (1956)



A group of young scientists met at Dartmouth College (USA) to have a 2-month brainstorming session “to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves”.

John McCarthy coined the term “Artificial Intelligence”.

Over to you!

The scientists who took part in the Dartmouth Workshop were a group of men from one specific background (mostly US/UK mathematicians).

If you were going to hold a “Summer Workshop” today to define the future of AI, who would you invite and why?

What is AI?

Artificial Intelligence is the ability of computers to perform tasks that typically require human intelligence.

These are tasks that involve processes such as learning, thinking, understanding, reasoning, problem-solving, decision-making, inference, identifying objects, generating language, etc.

The rule-based era (1960–1990)

Symbolic AI

Solutions were designed using clearly defined algorithms, heuristics, logical representations or IF-THEN rules.

Examples: rule-based systems and search algorithms.

The knowledge is coded in the *algorithm*.

```
<rule>
  <pattern>
    <token regexp="yes">what|which</token>
    <token postag="JJ.*|NN.*" postag_regexp="yes" min="0" />
    <token postag="NN.*" postag_regexp="yes" chunk_re="E-NP.*" />
    <token postag="MD">
      <exception>need</exception>
    </token>
    <token inflected="yes" min="0">not</token>
    <token
  regexp="yes">s?he|it|I|we|they|you|this|that|th[oe]se| (some|any|every|no) (thing|bo
  dy|one)</token>
    <token postag="RB" min="0">
      <exception postag="IN|EX" postag_regexp="yes" />
      <exception regexp="yes">live|close|last</exception>
    </token>
    <marker>
      <token postag="VB[GDNZ]" postag_regexp="yes" skip="-1">
        <exception postag="VB" />
        <exception regexp="yes">kids|physics|knights|let|fit</exception>
        <exception>'s</exception>
      </token>
    </marker>
    <token>?</token>
  </pattern>
  <message>The correct verb form after the modal verb "\4" is <suggestion><match
  no="8" postag="VB" /></suggestion>.</message>
  <example correction="buy">Which car would you
<marker>bought</marker>?</example>
  <example correction="choose">Which one would you <marker>chose</marker> and
  why?</example>
</rule>
```

A grammar rule from LanguageTool.

ELIZA: The First Chatbot (1966)

ELIZA was a simple program created at the MIT that mimicked a therapist.

It worked by flipping the user's questions back at them using predefined rules.

```
Welcome to

EEEEEE LL      IIII  ZZZZZZ  AAAAA
EE      LL      II    ZZ    AA  AA
EEEEEE LL      II    ZZ    AAAAAA
EE      LL      II    ZZ    AA  AA
EEEEEE LLLLLL IIII ZZZZZZ  AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```



The AI Winter (late 1970s)

AI faced strong criticism, as many scientists believed AI research had little real-world value and was something impossible to achieve.

Computers were too slow, and rules were too rigid to handle the messiness of the real world.

As many projects failed to deliver their promises, funding for AI research dried up.



Expert Systems (1980s)

Expert systems were big rule-based systems that attempted to mimic the human decision-making and reasoning of experts in a particular domain, e.g. medicine.

To build them, experts had to explain their thinking step by step. This process, called knowledge engineering, was laborious and time-consuming. The systems couldn't handle new situations or learn from their mistakes.

Over to you!

Can you speak a new language only by reading a dictionary and a grammar book? Why/why not?

More Flexible Approaches

Researchers explored methods that could handle uncertainty and learn from experience.

Examples:

- Fuzzy logic
- Probabilistic methods
- Genetic algorithms
- Neural Networks

There was a shift towards statistical methods.

The data age (1990–2010)

Data-driven AI: Machine Learning (ML)

Machines can “learn” from examples instead of descriptions.
E.g. don't tell the machine what a cat is; show it a million cats.

The success of ML was due to the increasing availability of big volumes of (training) data.

The knowledge is in the *data* (we don't need domain experts any more).

ML can solve virtually any problem (and more accurately than before).

Machine Learning Algorithms

Some well-known algorithms:

- Logistic Regression
- Neural Networks
- Support Vector Machines
- Random Forests
- Conditional Random Fields
- K-Nearest Neighbours

Some tasks that use ML:

- Automatic Speech Recognition
- Text to speech
- Machine translation
- Sentiment analysis
- Image recognition
- Recommendation systems
- ...

The power of Deep Learning (2010–2017)

Deep Learning

- Big neural networks, with more than two hidden layers.
- Adding more layers increases their expressive power but also requires massive volumes of data and more powerful computational resources (GPUs).
- These networks can identify what is important in the input all by themselves – no need to explicitly define “features”.
- Many of these models are “end-to-end”, i.e. they can learn to predict values from the input without explicit intermediate representations.
- Machines gain the ability to “understand” meaning.
- Greater abstraction capabilities and performance.
- Huge advances in tasks such as machine translation, image processing, summarisation, etc.

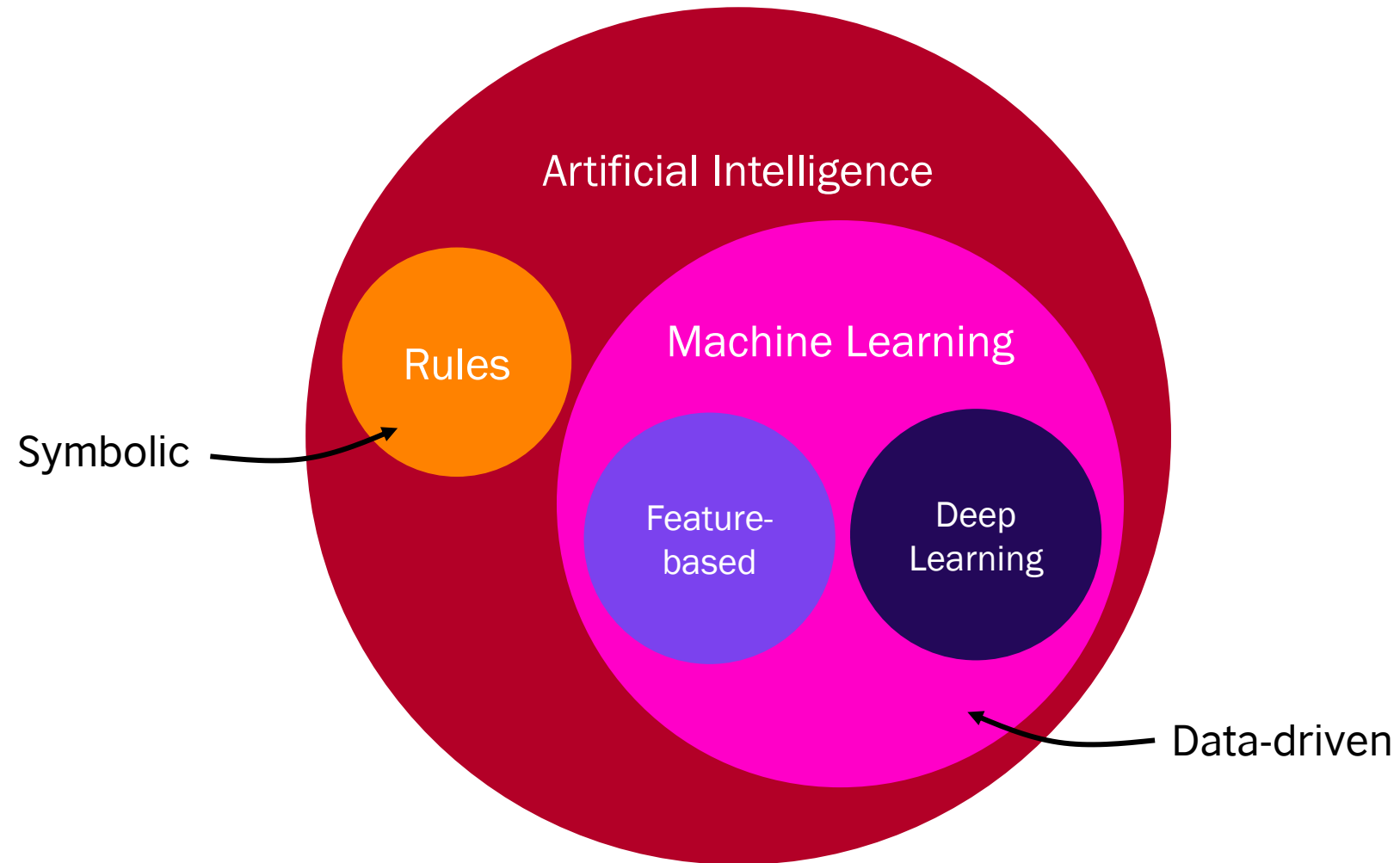
The Generative AI revolution (2017–today)

Generative AI



- The emergence of new powerful neural networks (known as “Transformers”) gave machines greater understanding capabilities.
- Large Language Models such as ChatGPT (2022) revolutionised the field and led to an unprecedented interest and “democratisation” of AI.
- AI moved from “recognition” to “generation”.
- Anyone can generate text, images, video, sound, etc. with just a “prompt” (plain text instructions) – no need to code or be an AI expert.
- The quality of the results depend on the quality of the prompt (“prompt engineering”).
- Results are not perfect – they may contain hallucinations and inaccuracies.

Summary of AI approaches

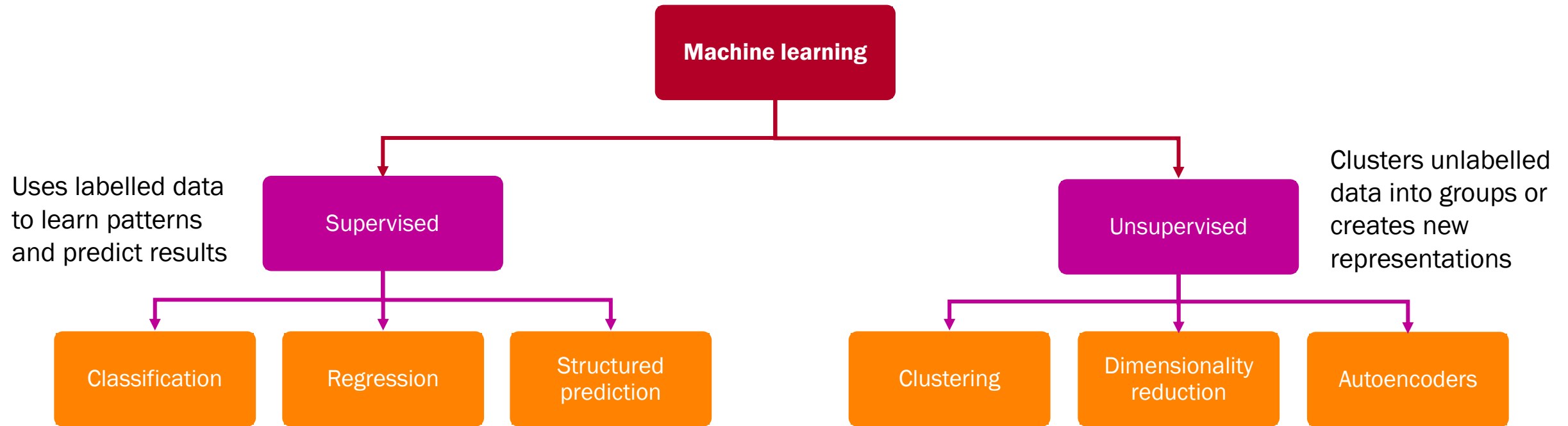




How does modern AI work?

How does modern AI work?

Machine Learning



Supervised learning

Regression

Learn to predict a numerical value (discrete or continuous).

EXAMPLES

- Score a piece of writing automatically.
- Predict stock prices.
- Predict temperature.
- Estimate sales.

Supervised learning

Classification

- Assign a class to an instance/data point.
- Classification can be binary or multi-class.

EXAMPLES

Binary

- Spam detection.
- Pass/fail an exam.
- Test positive/negative for a health condition.

Multiclass

- Image classification.
- Classify students by language proficiency level.
- Classify dogs into breeds.

Supervised learning

Structured prediction

Predict structured output, i.e. sequences.

EXAMPLES

- Part-of-speech tagging.

This is a nice house $\Rightarrow$ This/PRON is/VERB a/DET nice/ADJ house/NOUN

- Optical character recognition.

My car is sick $\Rightarrow$ My cat is sick

- Machine translation.

Il libro è interessante. $\Rightarrow$ The book is interesting.

Unsupervised learning

Clustering

Group instances into classes based on their similarity.

EXAMPLES

- Group customers by purchasing habits.
- Group articles by topics.
- Segment areas based on criminal activity.

Unsupervised learning

Dimensionality reduction

Reduce a large collection of values, such as a vector of features, into a more compact representation containing the most salient features.

EXAMPLE

A vector containing 1,000 values that represent different aspects of a word (length, absolute frequency, relative frequency, POS, number of times it follows a preposition, number of times it follows a noun, etc.)

[3.63, 657, 269.1, 63.36, 8, 0.4536, 13.5, ..., 245.134, 25.17, 0.9, 6.72, 0.92, 0.0001, 52]



[0.3, 0.536, 3223, 335.8, 1, 145, 83.6, ..., 829.5]

A new vector of 100 dimensions that contains the most important information.

Unsupervised learning

Autoencoders

The representation of an instance as a vector of numerical values.

EXAMPLE

Representing a sentence as a vector of numbers.

Monkeys are agile,
intelligent primates. → [25.32, 6.2, 2.534, 0.23, 0.004, 682, ..., 0.53, 3.758]

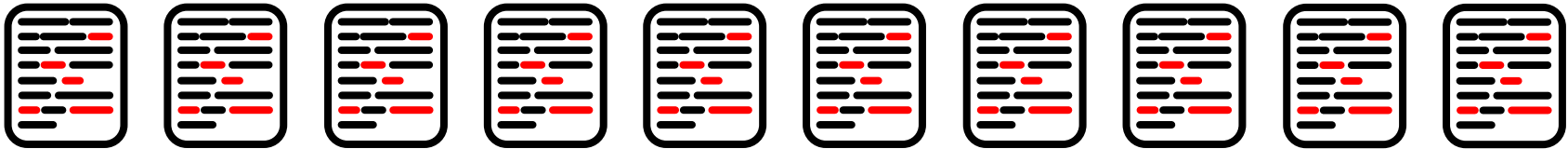


How does supervised machine learning work?



Data collection

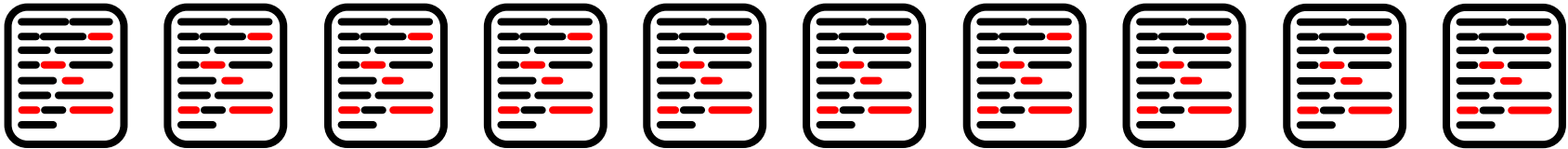
Examples





Data collection

Examples

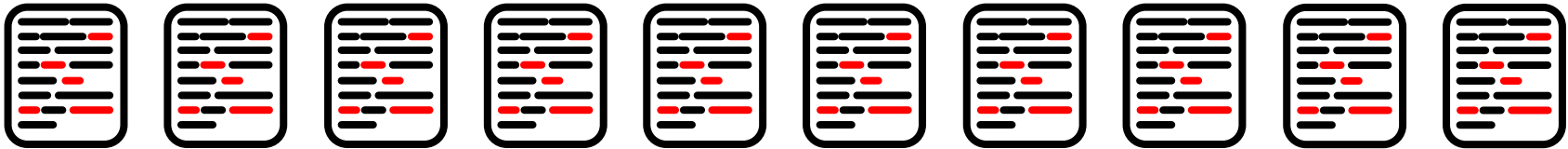


Labels

2 3 8 2 9 3 7 7 10 4

Data collection

Examples

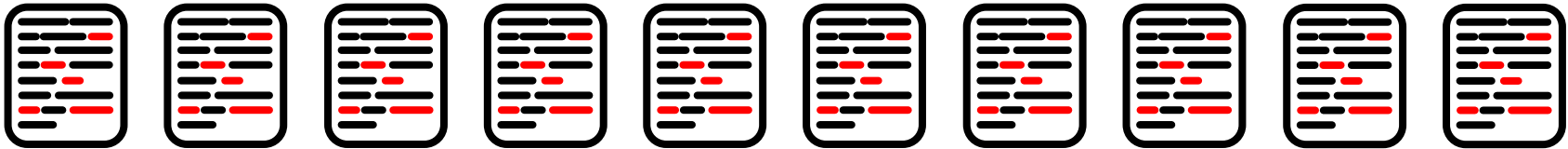


Labels

GOOD BAD GOOD BAD BAD GOOD GOOD GOOD GOOD BAD

Data collection

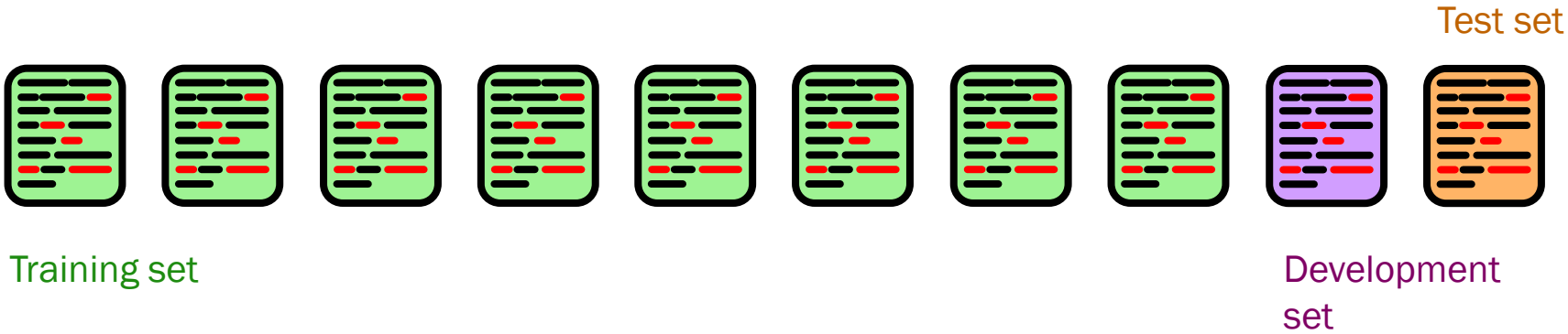
Examples



Labels

EASY MEDIUM MEDIUM EASY HARD EASY HARD HARD HARD HARD MEDIUM

Data collection



Training

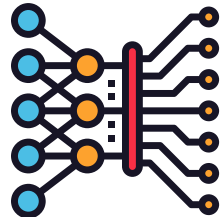


Training
data

Training



Training
data

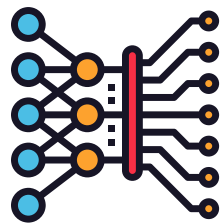


Machine
learning

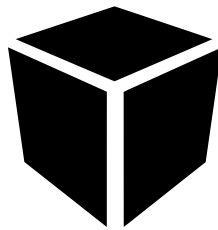
Training



Training
data



Machine
learning

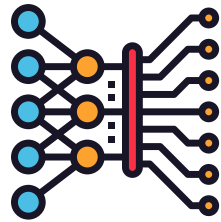


Model

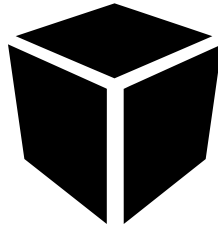
Fine-tuning



Training
data

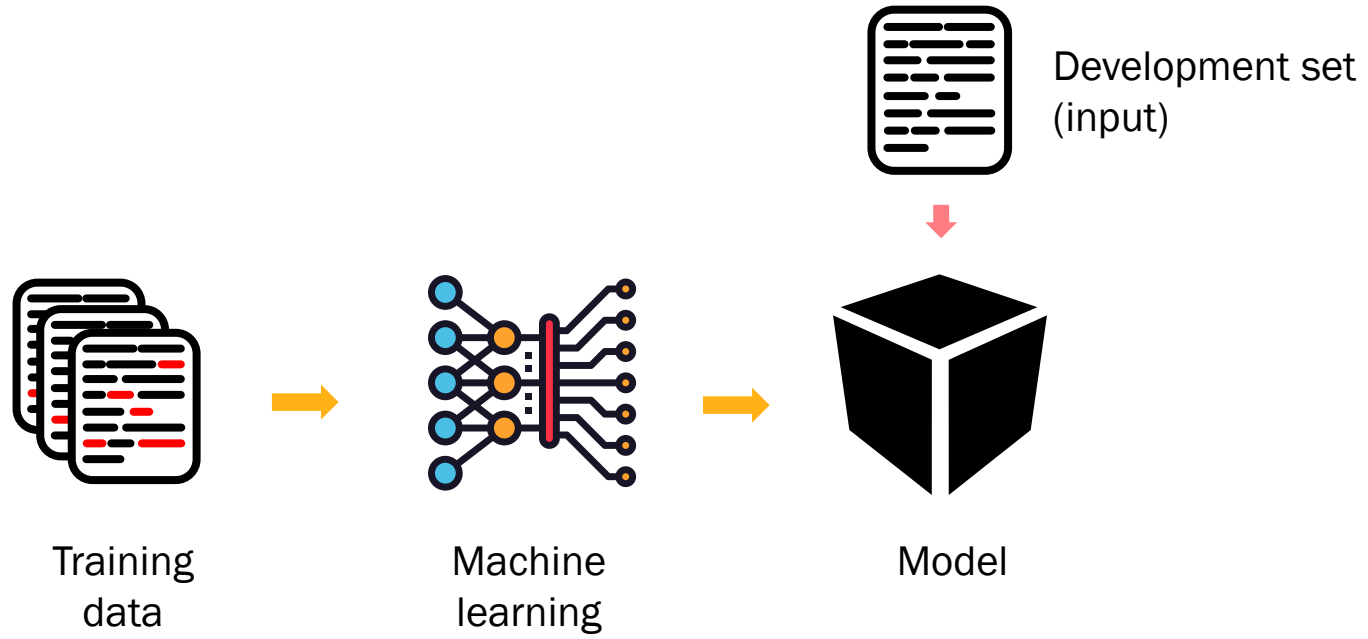


Machine
learning

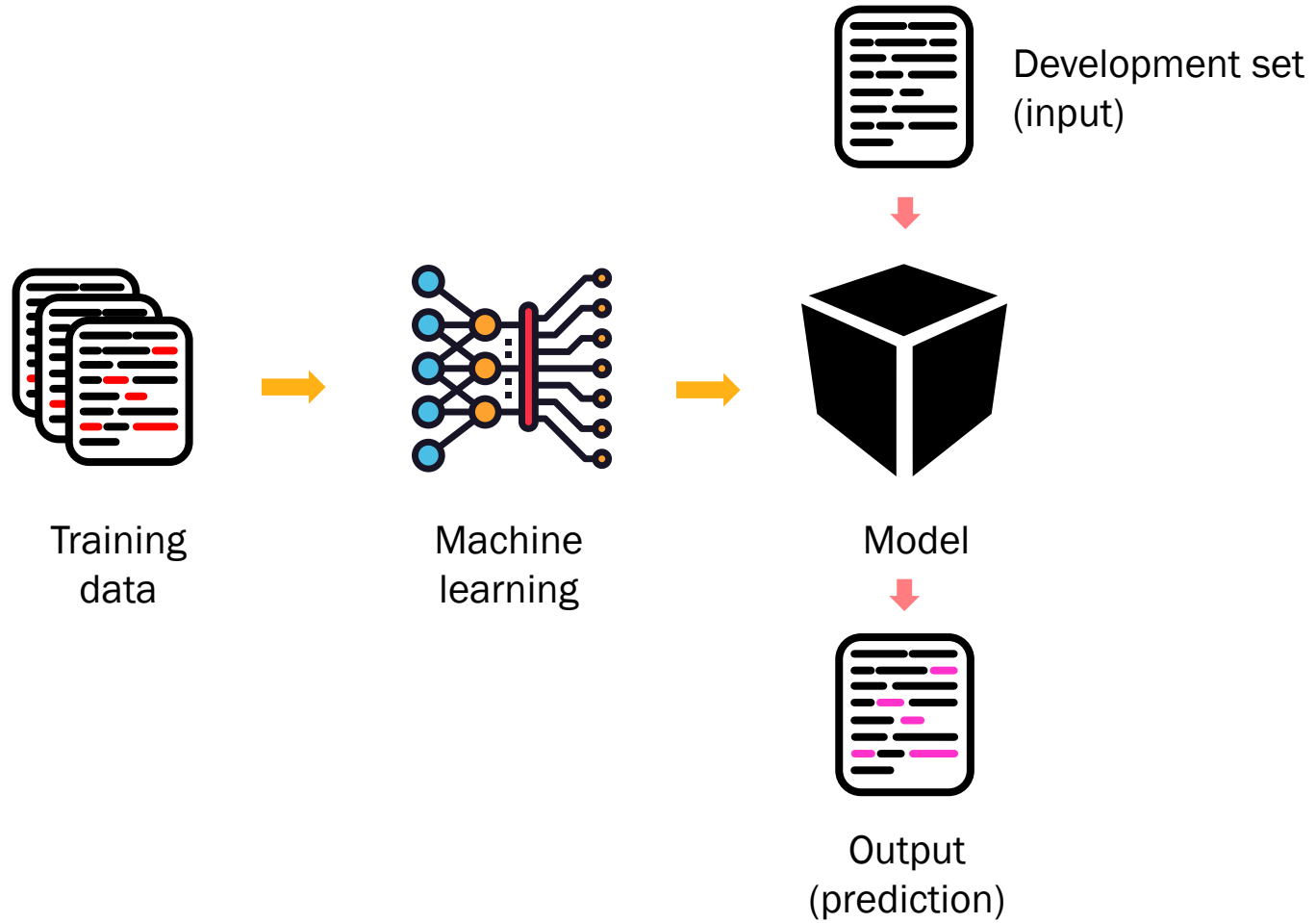


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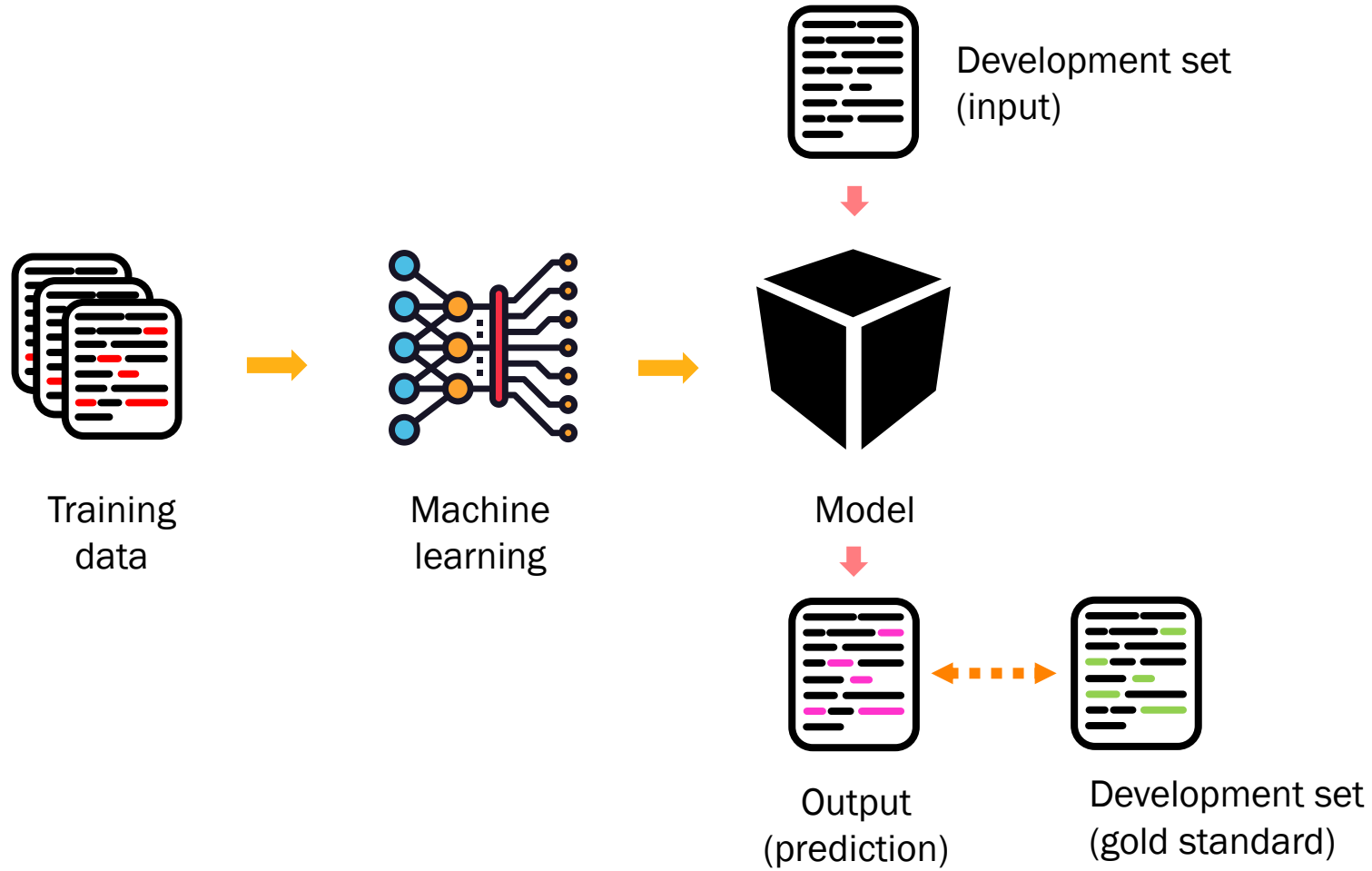
Fine-tuning



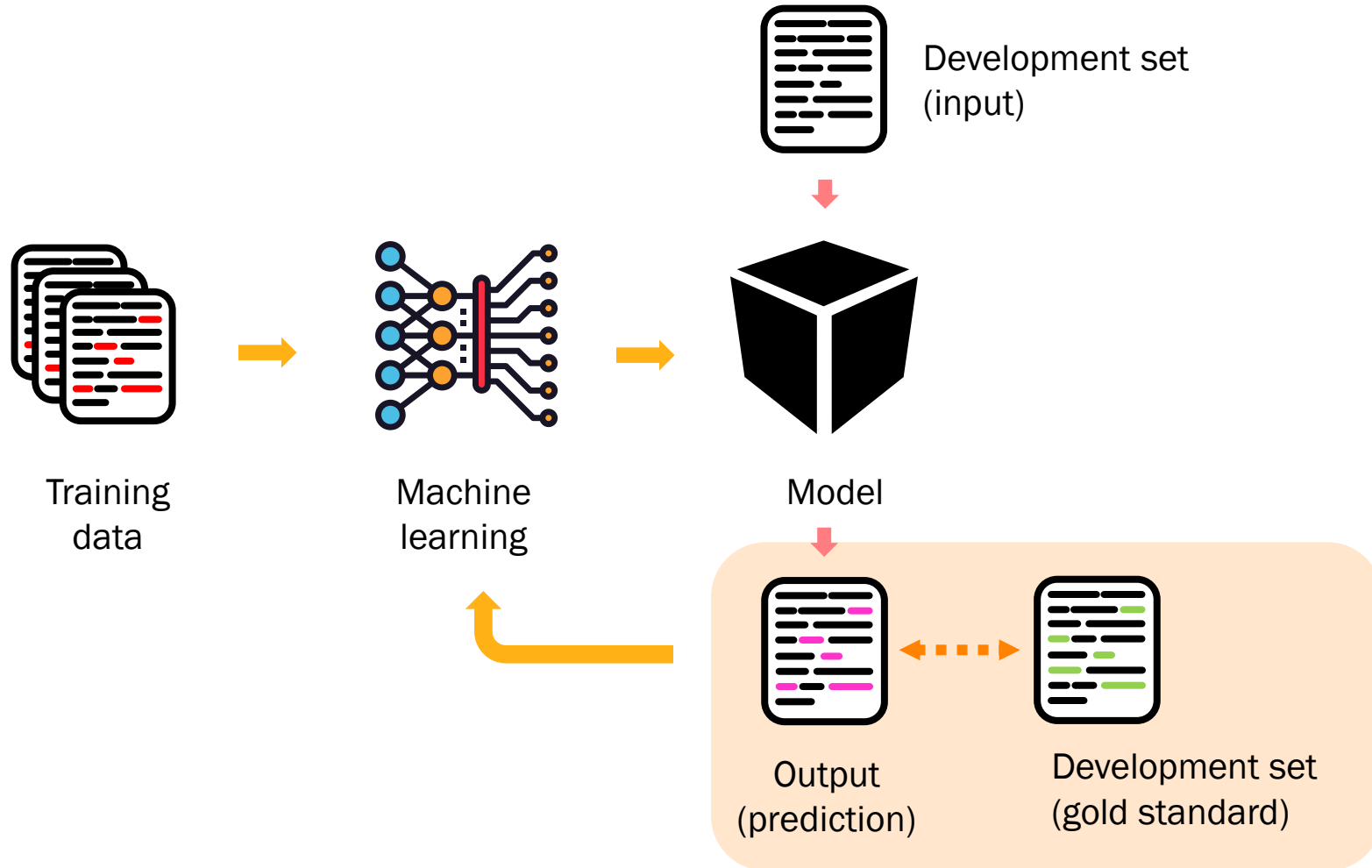
Fine-tuning



Fine-tuning



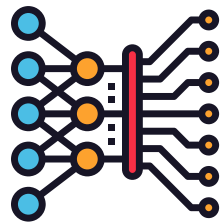
Fine-tuning



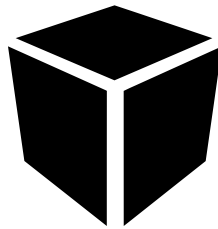
Testing



Training
data

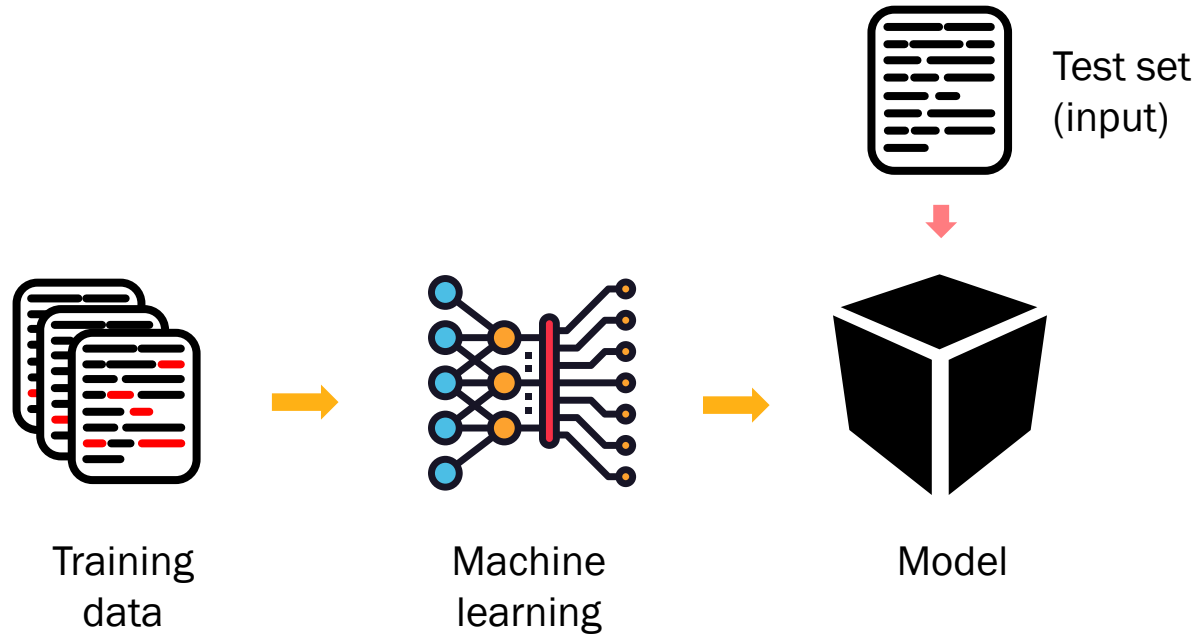


Machine
learning

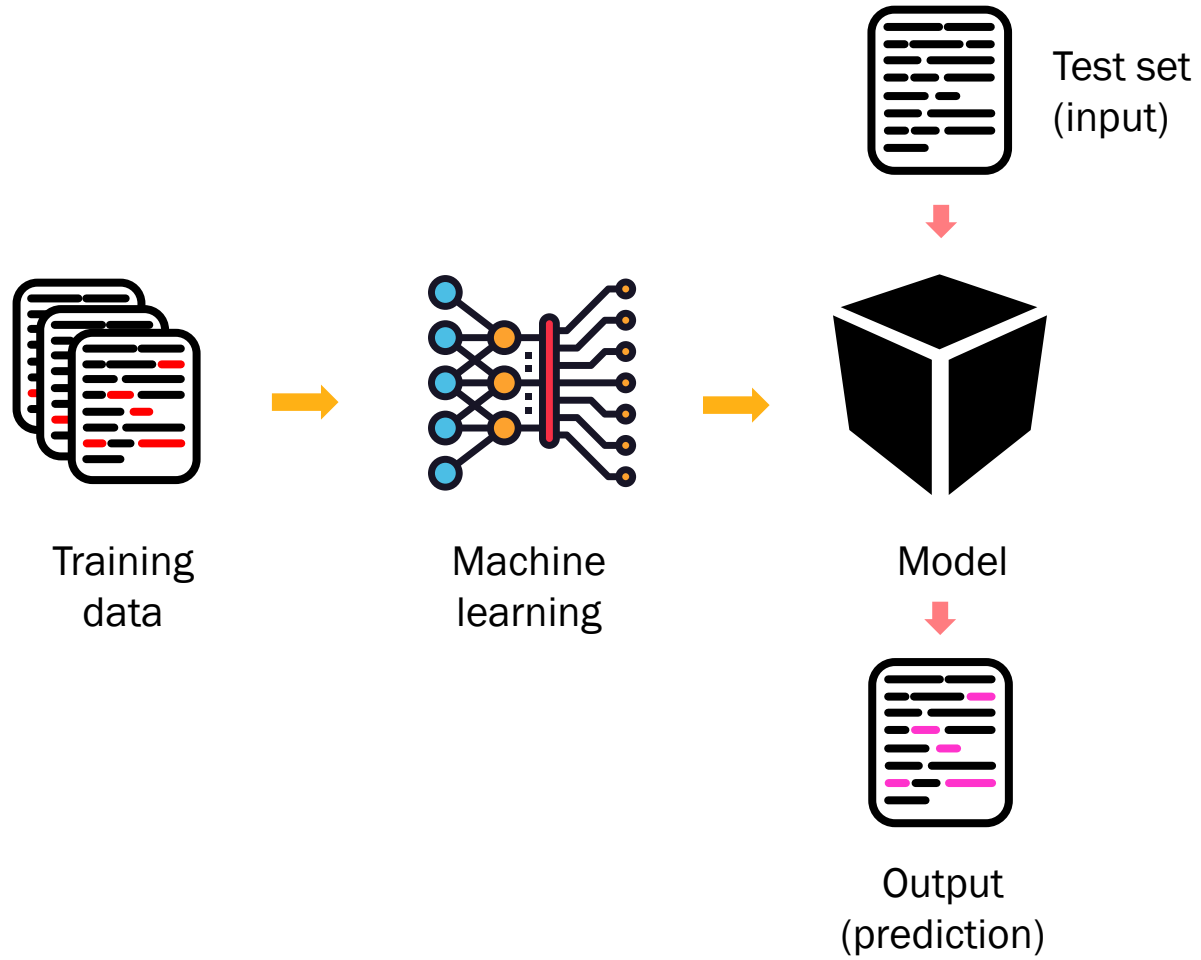


Model

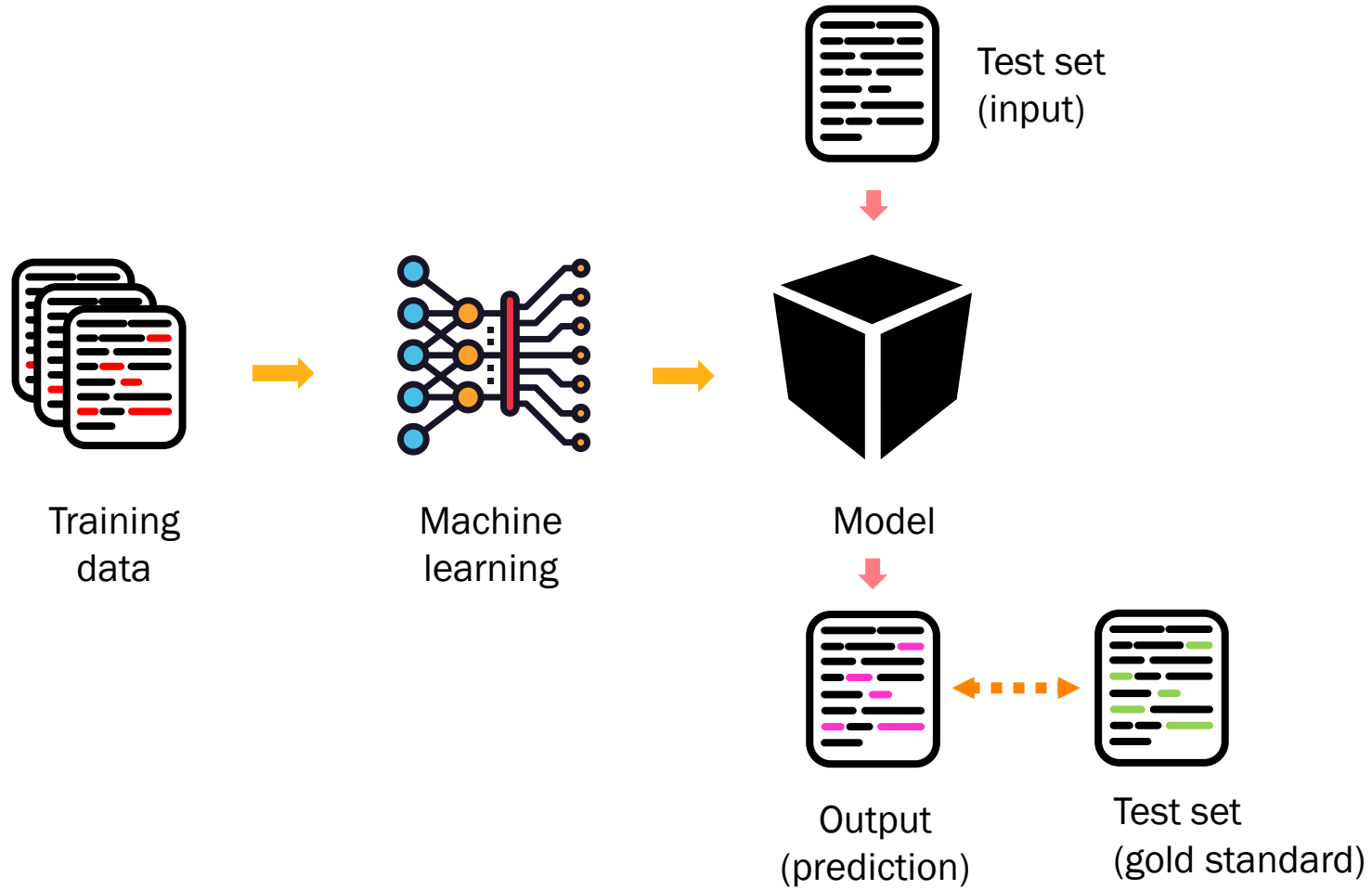
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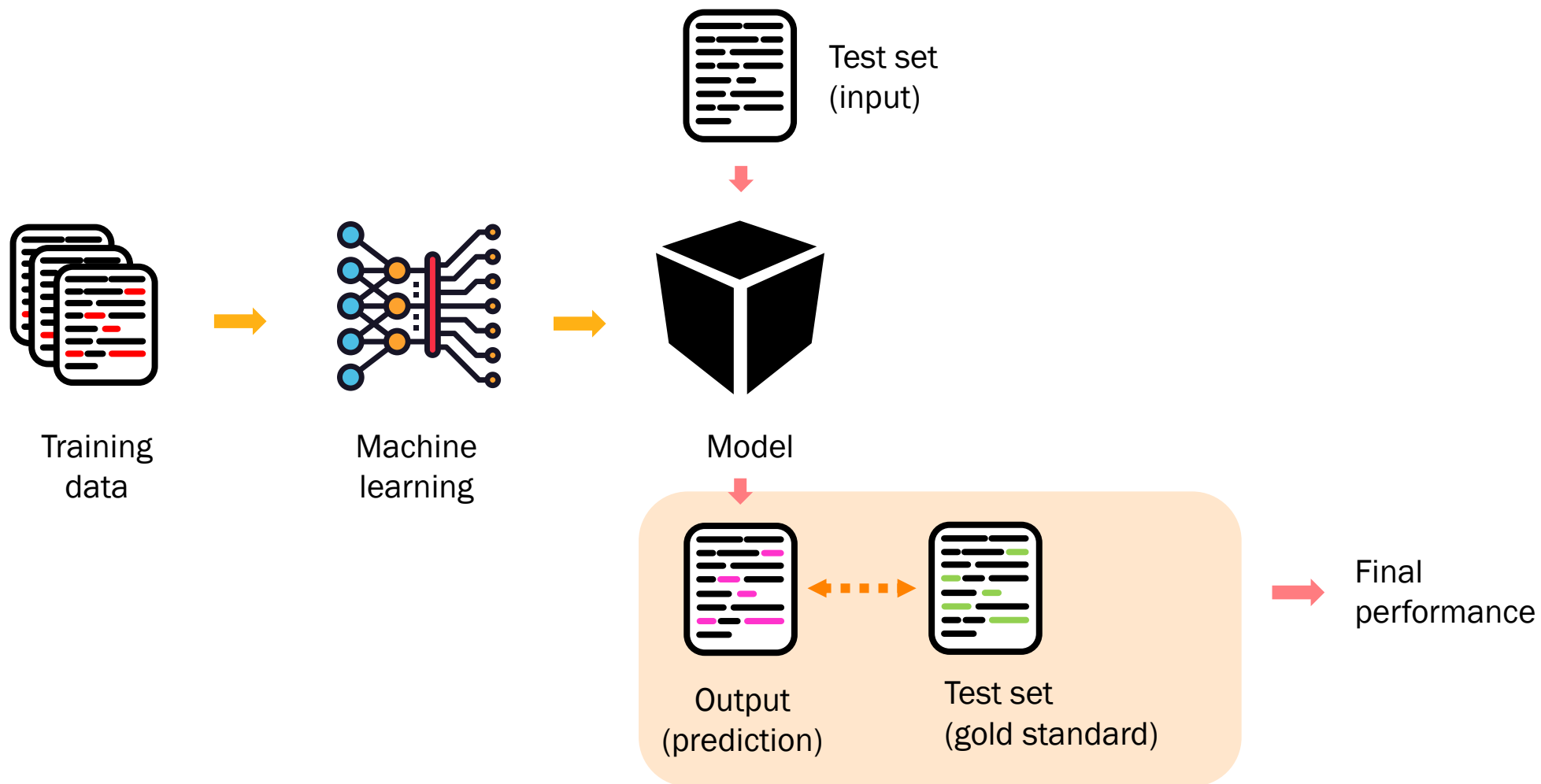
Testing



Testing



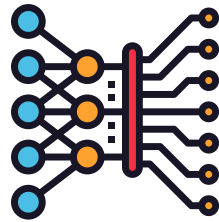
Testing



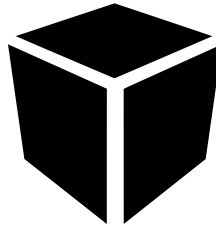
Real-world usage



Training
data

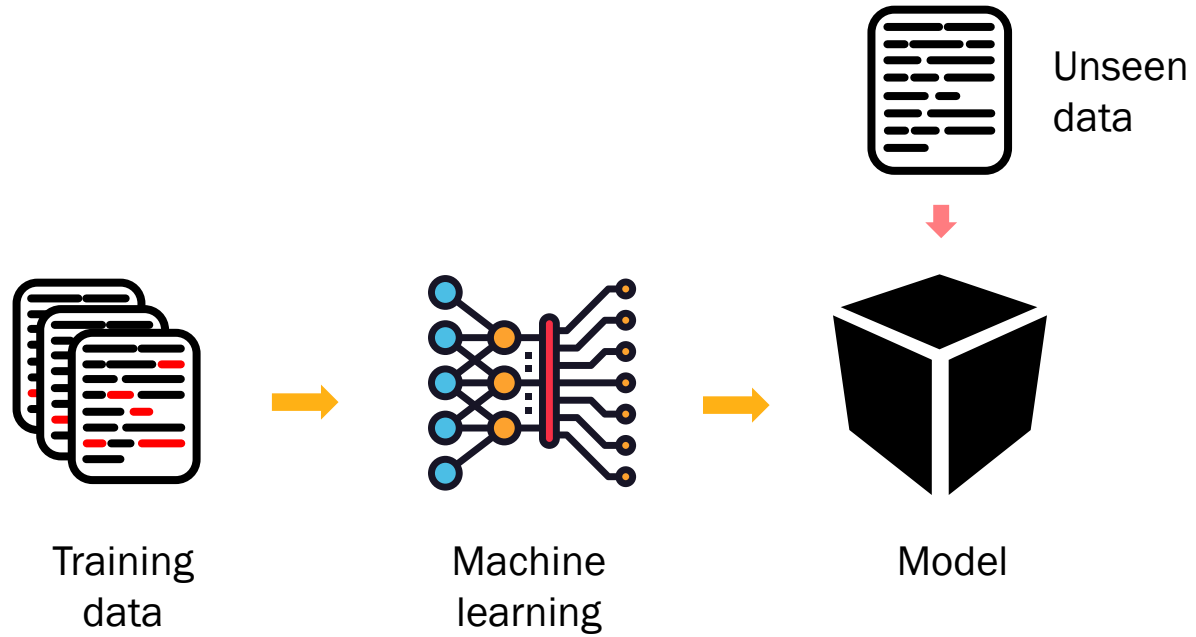


Machine
learning

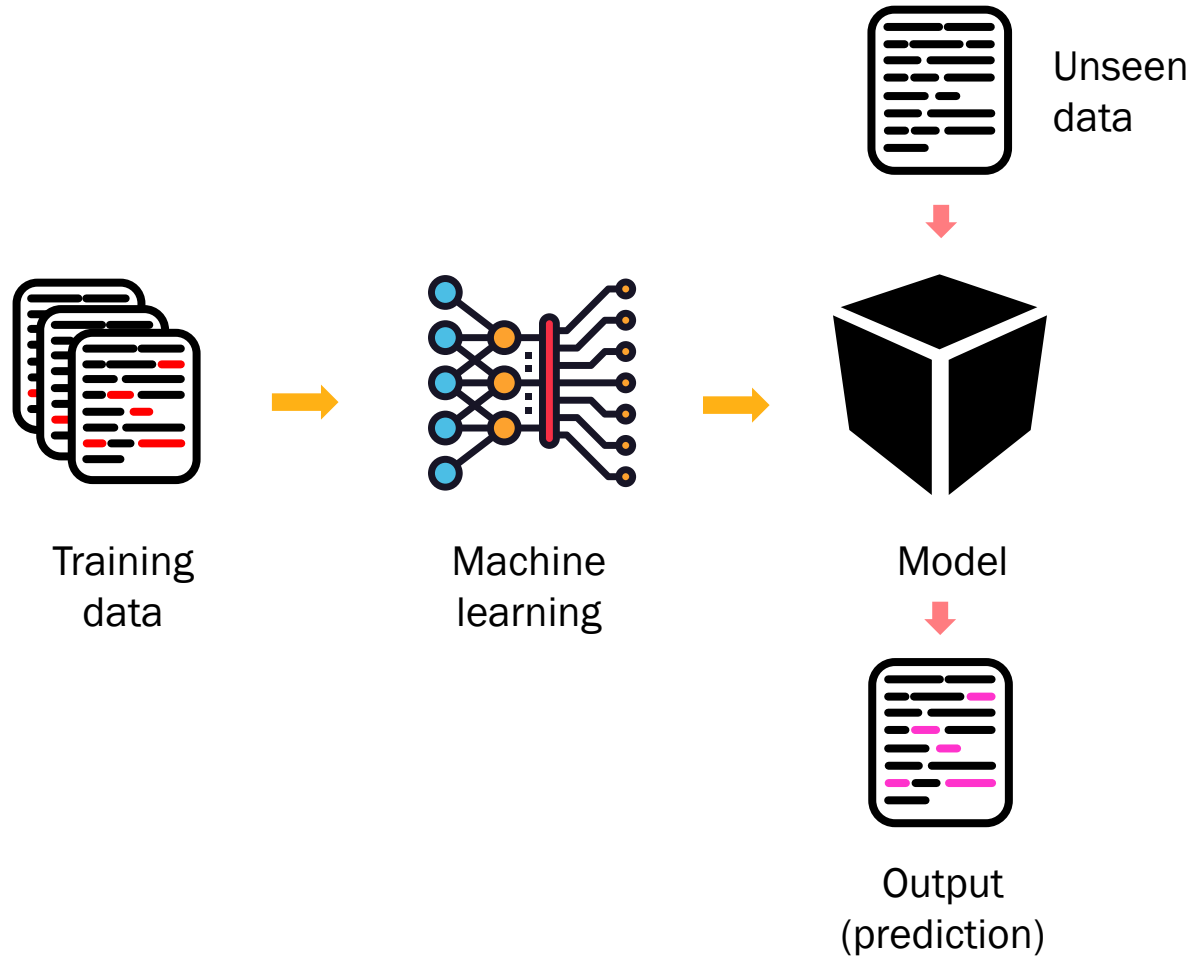


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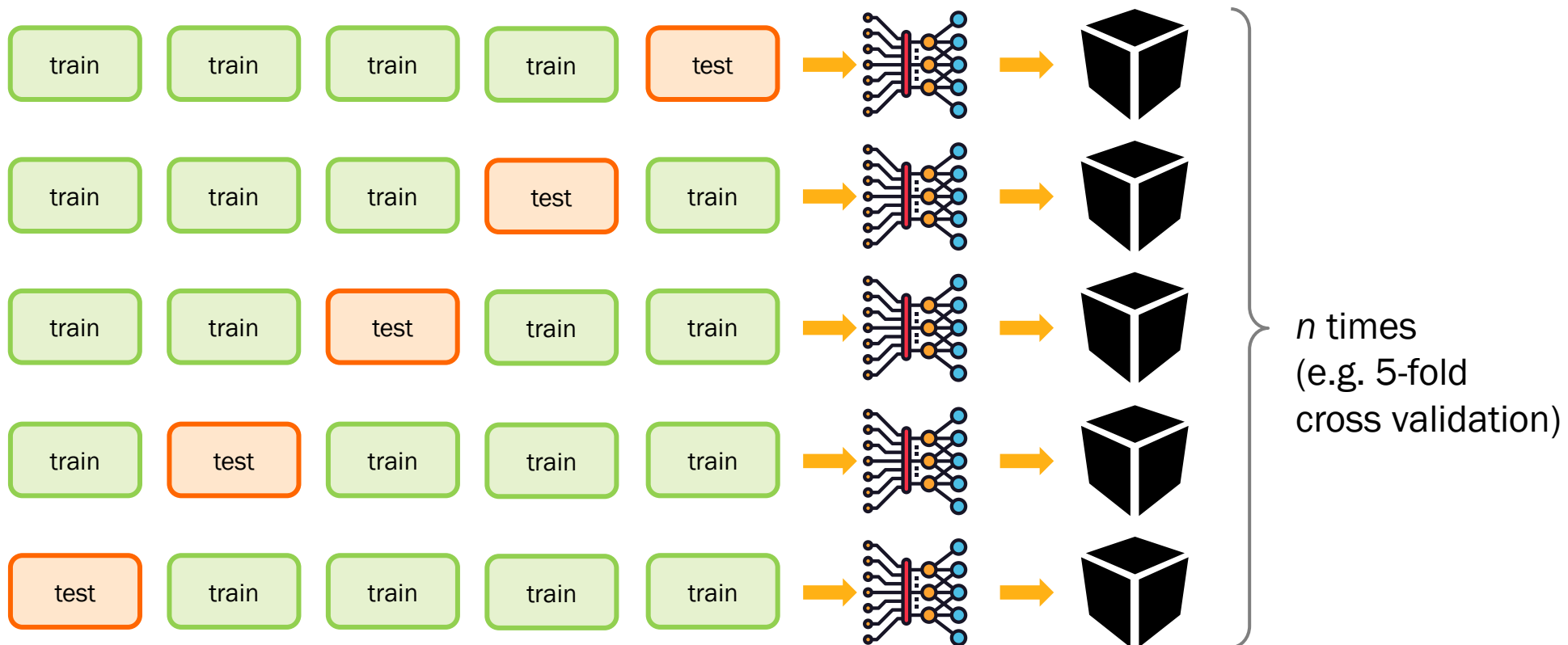
Real-world usage



Real-world usage



Cross validation



Feature-based models

Training instances are defined as a set of “features”, i.e. information about different aspects of the input that can help us predict the target value.

Example: Determine if a patient has COVID, flu, a cold or an allergy based on their symptoms.

What information about each patient and their symptoms would be useful?

Feature-based models

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Example: Determine if a patient has COVID, flu, a cold or an allergy based on their symptoms.

What information about each patient and their symptoms would be useful?

Age: Patient age (18 to 85 years)

Sex: Biological sex (Female, Male)

Preexisting Condition: Medical background (None, Asthma, Diabetes)

Primary Symptom: Chief complaint (Sneezing, Runny Nose, Dry Cough, Shortness of Breath)

Temperature C: Body temperature (36.0 to 40.0 °C)

CRP mgL: Inflammatory marker (0.0 to 100.0+ mg/L)

SpO2 Percent: Blood oxygen (85.0 to 100.0 %)

Eosinophils Percent: Allergy biomarker (0.0 to 10.0 %)

Diagnosis: Target class (Allergy, Common Cold, COVID19, Influenza)

Our dataset

Age	Sex	Preexisting_Condition	Primary_Symptom	Temperature_C	CRP_mgL	SpO2_Percent	Eosinophils_Percent	Diagnosis
38	Female	None	Sneezing	36.7	0.8	98.6	7.1	Allergy
52	Male	Asthma	Dry_Cough	38.3	62.4	93.1	0.9	COVID19
29	Female	None	Runny_Nose		11.5	97.8	2.1	Common_Cold
61	Male	Diabetes	Dry_Cough	38.9	48.2	95.2	1.1	Influenza
44	Female	None	Sneezing	36.9	1.2	98.4	6.2	Allergy
33	Male	None	Shortness_of_Breath	38.5	58.1	92.4	0.8	COVID19
57	Female	Asthma	Runny_Nose	37.6	13.8	97.2	1.8	Common_Cold
48	Male	None	Dry_Cough	999	999	96.1	1.3	Influenza
22	Female	None	Sneezing		0.5	99.1	6.8	Allergy
65	Male	Diabetes	Shortness_of_Breath	38.2	68.3	91.8	1.0	COVID19
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Preprocessing

We can't use the data “as is” – we have to prepare it for machine learning.

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Preprocessing

We may need to fix any problems in our data, such as missing values or corrupt records.

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22	Female	None	Sneezing		0.5	99.1	6.8	Allergy
65	Male	Diabetes	Shortness_of_Breath	38.2	68.3	91.8	1.0	COVID19
41	Female	None	Runny_Nose	37.3	9.8	98.0	2.4	Common_Cold
50	Male	Asthma	Dry_Cough	38.4	49.5	95.8	1.2	Influenza

Preprocessing

We may need to fix any problems in our data, such as missing values or corrupt records.

Age	Sex	Preexisting_Condition	Primary_Symptom	Temperature_C	CRP_mgL	SpO2_Percent	Eosinophils_Percent	Diagnosis
38	Female	None	Sneezing	36.7	0.8	98.6	7.1	Allergy
52	Male	Asthma	Dry_Cough	38.3	62.4	93.1	0.9	COVID19
29	Female	None	Runny_Nose	37.9	11.5	97.8	2.1	Common_Cold
61	Male	Diabetes	Dry_Cough	38.9	48.2	95.2	1.1	Influenza
44	Female	None	Sneezing	36.9	1.2	98.4	6.2	Allergy
33	Male	None	Shortness_of_Breath	38.5	58.1	92.4	0.8	COVID19
57	Female	Asthma	Runny_Nose	37.6	13.8	97.2	1.8	Common_Cold
48	Male	None	Dry_Cough	999	999	96.1	1.3	Influenza
22	Female	None	Sneezing	37.9	0.5	99.1	6.8	Allergy
65	Male	Diabetes	Shortness_of_Breath	38.2	68.3	91.8	1.0	COVID19
41	Female	None	Runny_Nose	37.3	9.8	98.0	2.4	Common_Cold
50	Male	Asthma	Dry_Cough	38.4	49.5	95.8	1.2	Influenza

Preprocessing

We may need to scale numerical ranges so that machine learning is more stable and converges more easily.

Age	Sex	Preexisting_Condition	Primary_Symptom	Temperature_C	CRP_mgL	SpO2_Percent	Eosinophils_Percent	Diagnosis
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Preprocessing

We may need to scale numerical ranges so that machine learning is more stable and converges more easily.

Age	Sex	Preexisting_Condition	Primary_Symptom	Temperature_C	CRP_mgL	SpO2_Percent	Eosinophils_Percent	Diagnosis
-0.455	Female	None	Sneezing	-1.332	-1.066	0.999	1.488	Allergy
0.639	Male	Asthma	Dry_Cough	0.725	1.258	-1.189	-0.73	COVID19
-1.159	Female	None	Runny_Nose	-1.203	-0.662	0.681	-0.301	Common_Cold
1.343	Male	Diabetes	Dry_Cough	1.496	0.722	-0.354	-0.658	Influenza
0.014	Female	None	Sneezing	-1.075	-1.051	0.919	1.166	Allergy
-0.846	Male	None	Shortness_of_Breath	0.982	1.096	-1.468	-0.765	COVID19
1.03	Female	Asthma	Runny_Nose	-0.175	-0.575	0.442	-0.408	Common_Cold
-1.706	Female	None	Sneezing	-1.203	-1.077	1.198	1.38	Allergy
1.656	Male	Diabetes	Shortness_of_Breath	0.597	1.481	-1.706	-0.694	COVID19
-0.22	Female	None	Runny_Nose	-0.561	-0.726	0.76	-0.194	Common_Cold
0.483	Male	Asthma	Dry_Cough	0.854	0.771	-0.115	-0.623	Influenza

Preprocessing

Machines only understand numbers, so we must convert all our data to a numerical representation.

Age	Sex	Preexisting_Condition	Primary_Symptom	Temperature_C	CRP_mg/L	SpO2_Percent	Eosinophils_Percent	Diagnosis
-0.455	Female	None	Sneezing	-1.332	-1.066	0.999	1.488	Allergy
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Preprocessing

Machines only understand numbers, so we must convert all our data to a numerical representation.

Age	Sex	Preexisting_Condition	Primary_Symptom	Temperature_C	CRP_mgL	SpO2_Percent	Eosinophils_Percent	Diagnosis
-0.455	0	0	0	-1.332	-1.066	0.999	1.488	0
0.639	1	1	1	0.725	1.258	-1.189	-0.73	1
-1.159	0	0	2	-1.203	-0.662	0.681	-0.301	2
1.343	1	2	1	1.496	0.722	-0.354	-0.658	3
0.014	0	0	0	-1.075	-1.051	0.919	1.166	0
-0.846	1	0	3	0.982	1.096	-1.468	-0.765	1
1.03	0	1	2	-0.175	-0.575	0.442	-0.408	2
-1.706	0	0	0	-1.203	-1.077	1.198	1.38	0
1.656	1	2	3	0.597	1.481	-1.706	-0.694	1
-0.22	0	0	2	-0.561	-0.726	0.76	-0.194	2
0.483	1	1	1	0.854	0.771	-0.115	-0.623	3

Let's code!

[s.id/css26](#)

Feature-based models

This is a multi-class classification problem.

Target variable: **Diagnosis** = {Allergy, Common Cold, COVID19, Influenza}

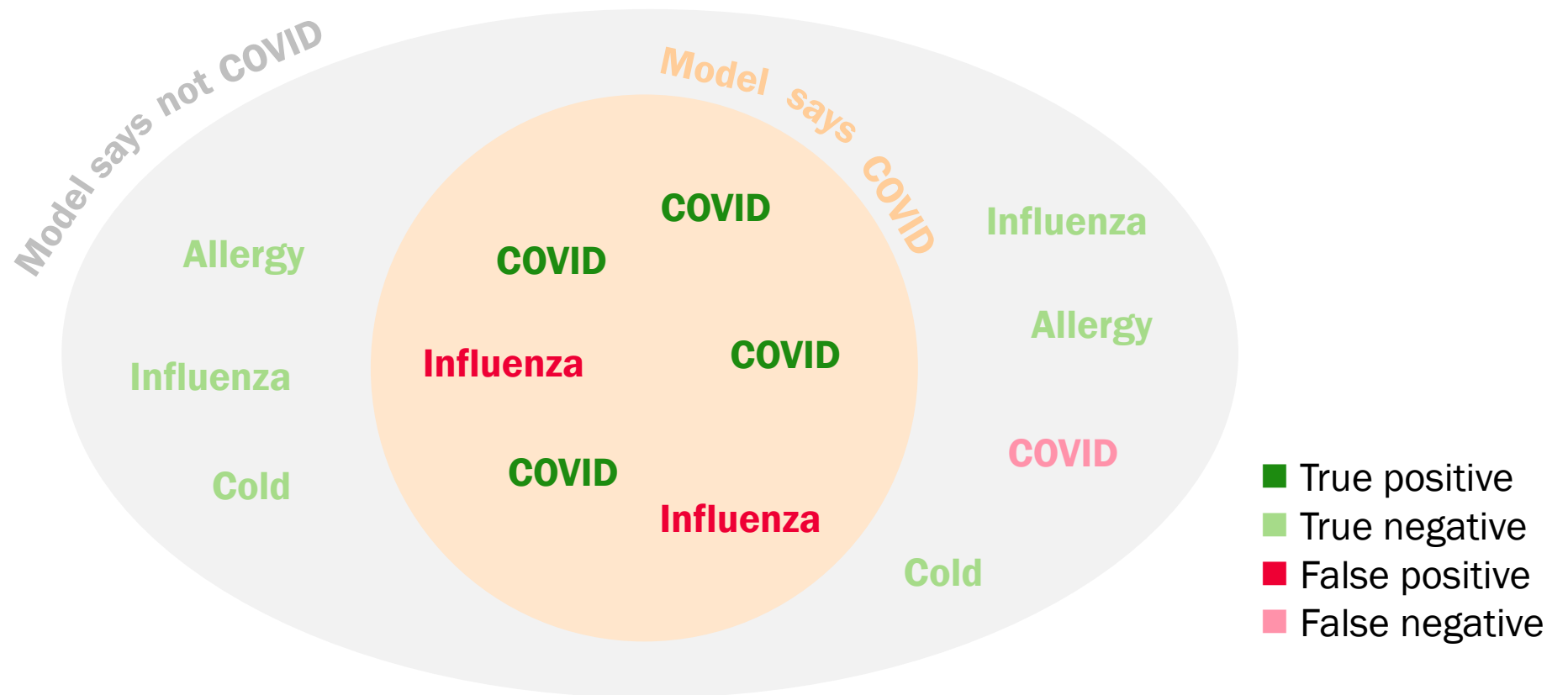
We can use different algorithms for training, e.g. Logistic Regression, Decision Trees, Random Forests, Neural Networks, Support Vector Machines, etc.

We train $\Rightarrow$ (optimise) $\Rightarrow$ test $\Rightarrow$ use.

Evaluation

Classification

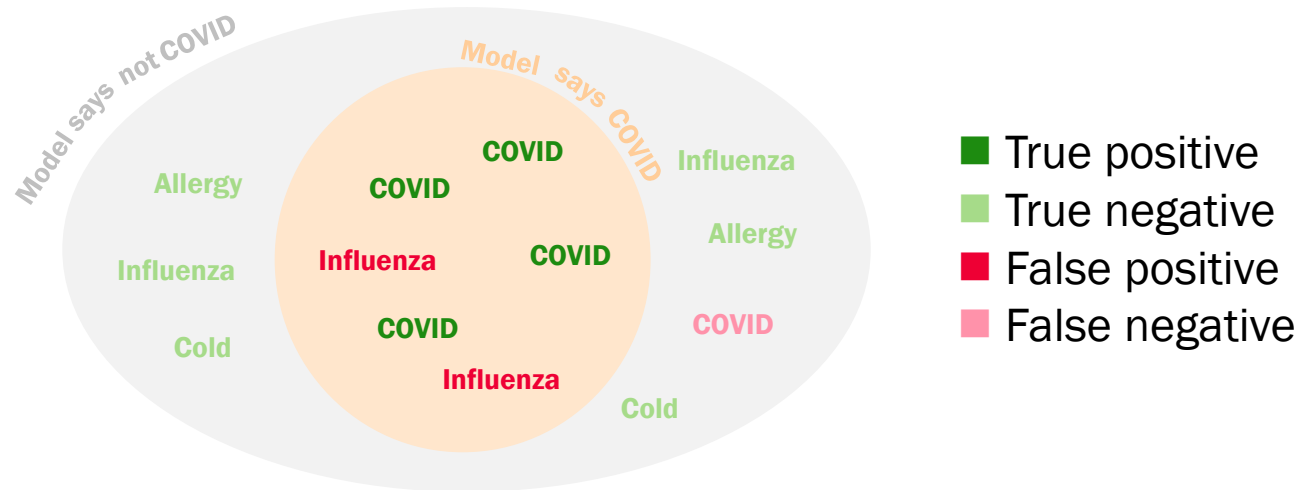
How do we know if our model is good? E.g. How well can we detect COVID?



Evaluation

Classification

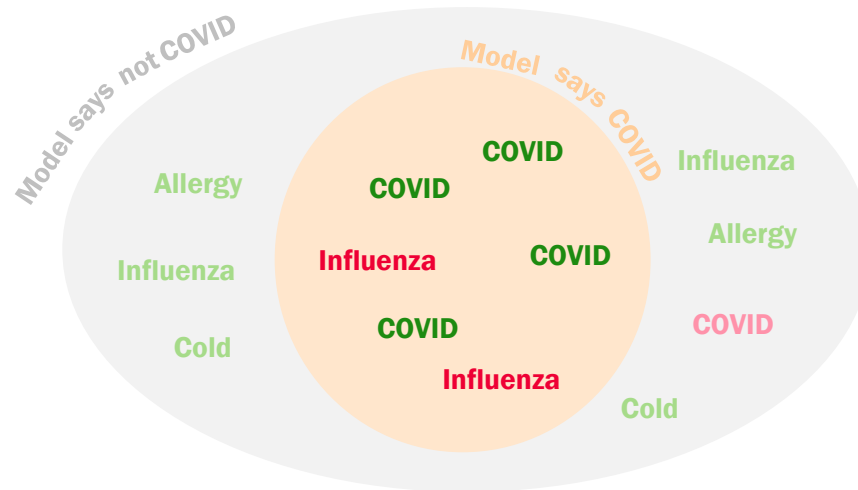
How well can we detect COVID?



Evaluation

Classification

How well can we detect COVID?



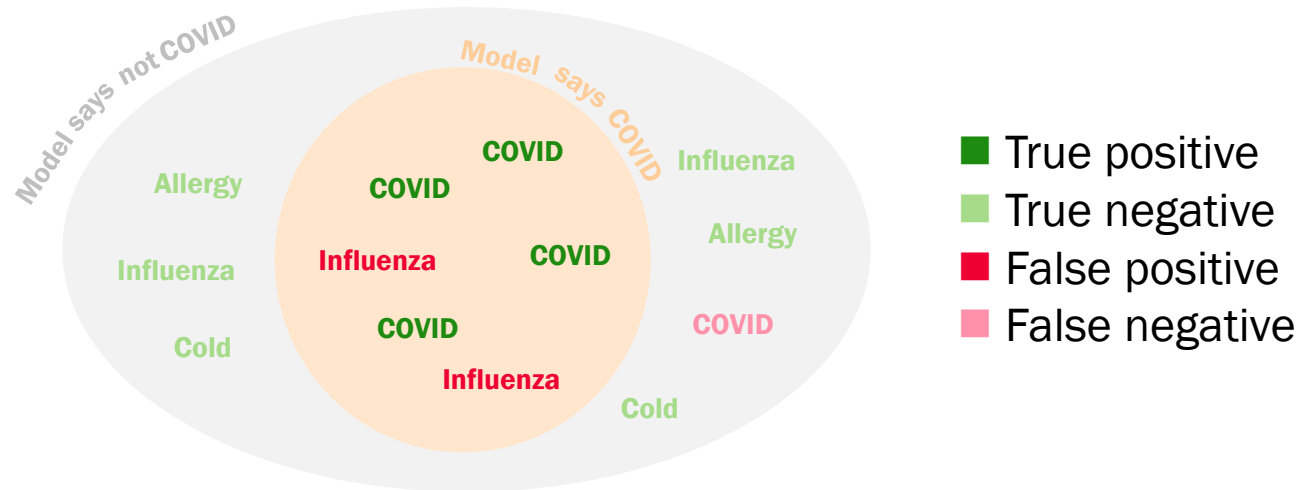
- True positive
- True negative
- False positive
- False negative

$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}}$$

Evaluation

Classification

How well can we detect COVID?

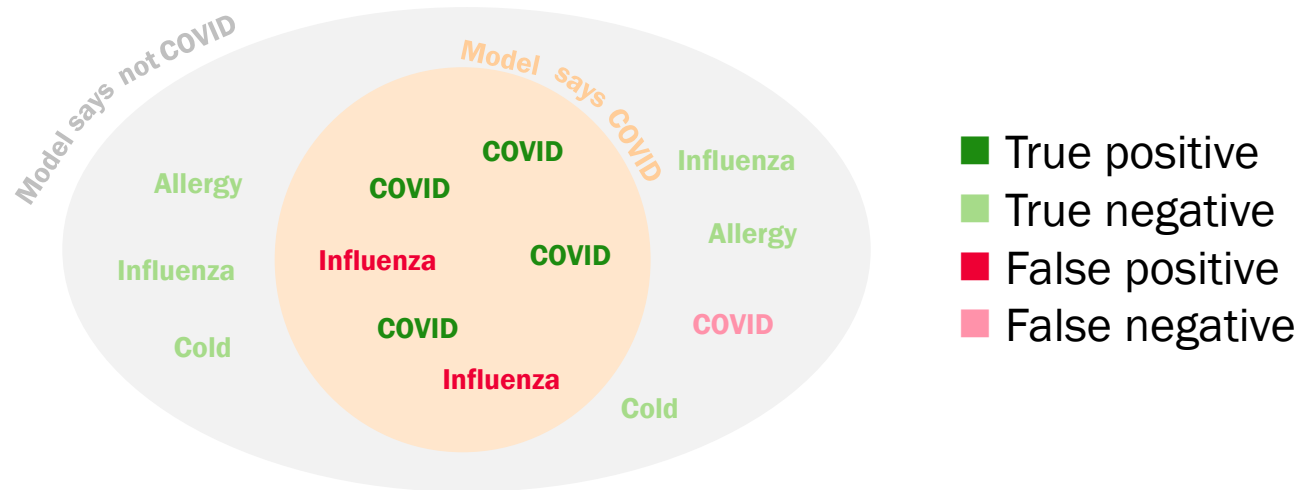


$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

Evaluation

Classification

How well can we detect COVID?



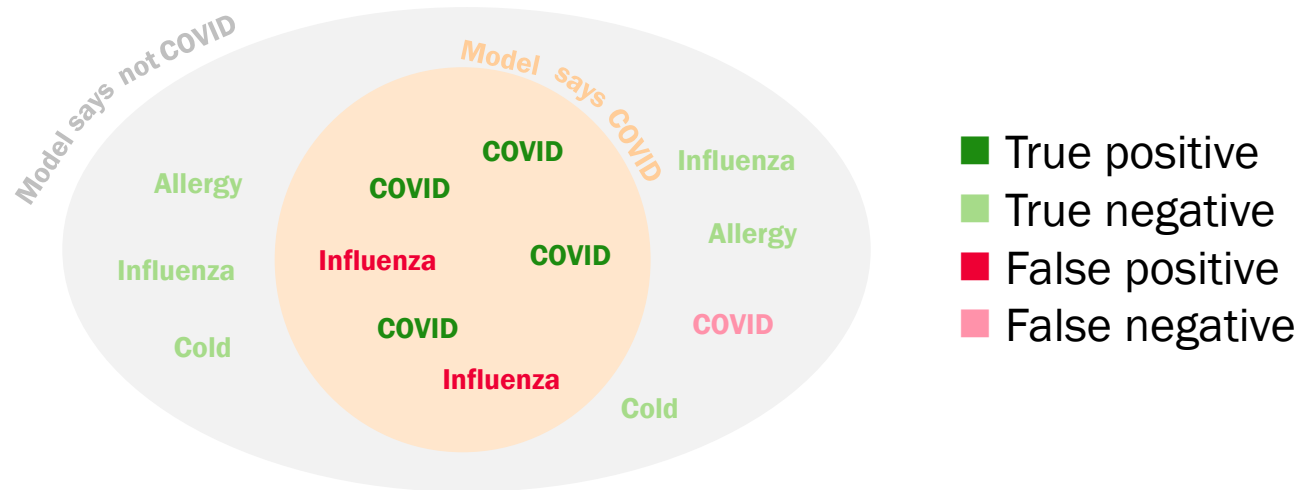
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$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}}$$

Evaluation

Classification

How well can we detect COVID?



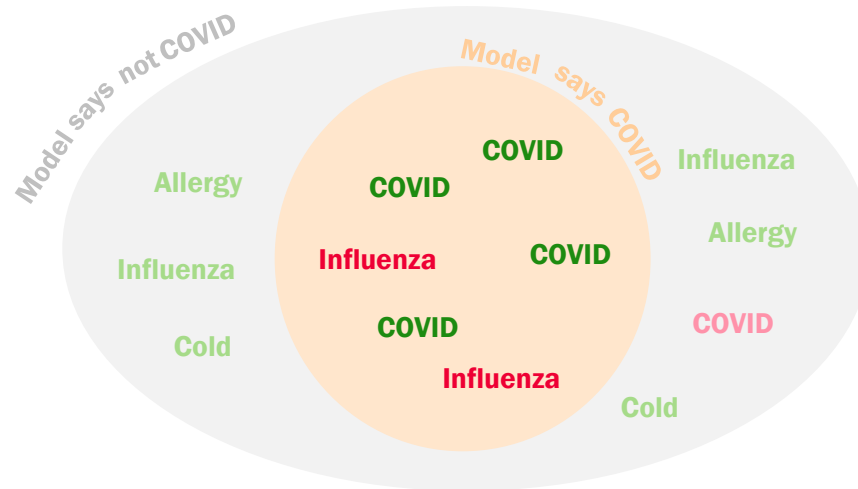
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$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

Evaluation

Classification

How well can we detect COVID?



- True positive
- True negative
- False positive
- False negative

$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

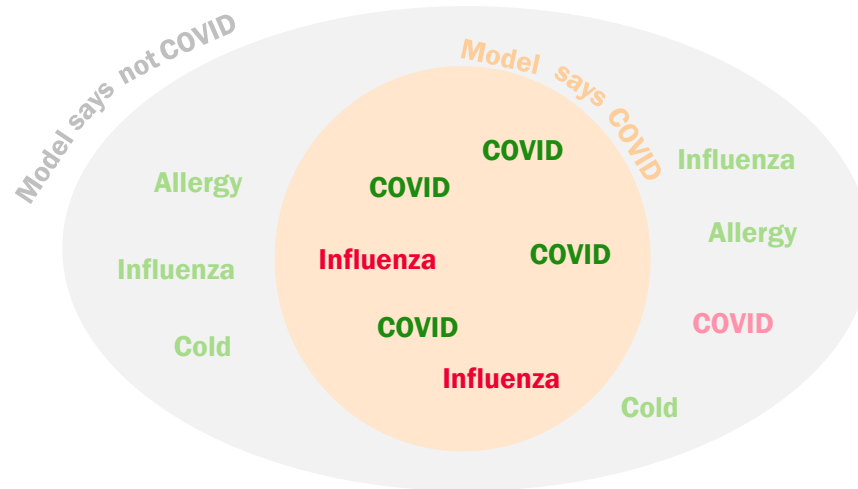
$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

$$\text{Recall} = \frac{\text{relevant and retrieved}}{\text{relevant}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}}$$

Evaluation

Classification

How well can we detect COVID?



$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

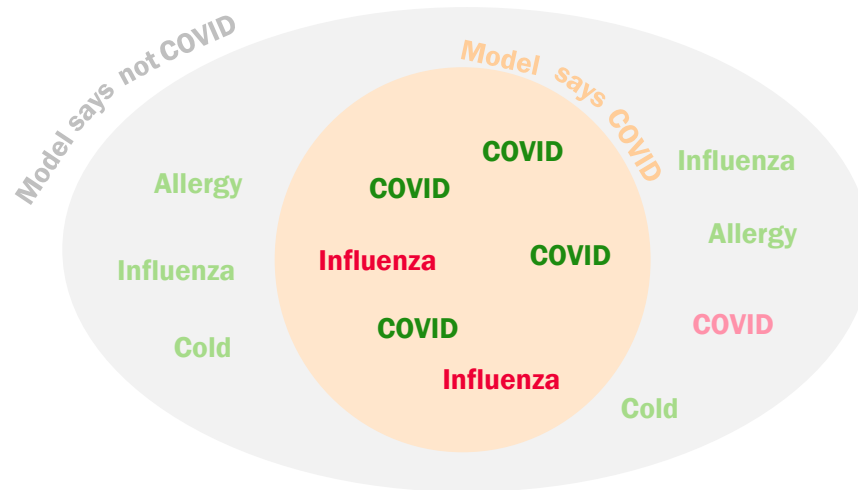
$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

$$\text{Recall} = \frac{\text{relevant and retrieved}}{\text{relevant}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}} = \frac{4}{4 + 1} = \frac{4}{5} = \mathbf{0.80}$$

Evaluation

Classification

How well can we detect COVID?



- True positive
- True negative
- False positive
- False negative

$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{■} + \text{■}}{\text{■} + \text{■} + \text{■} + \text{■}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{■}}{\text{■} + \text{■}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

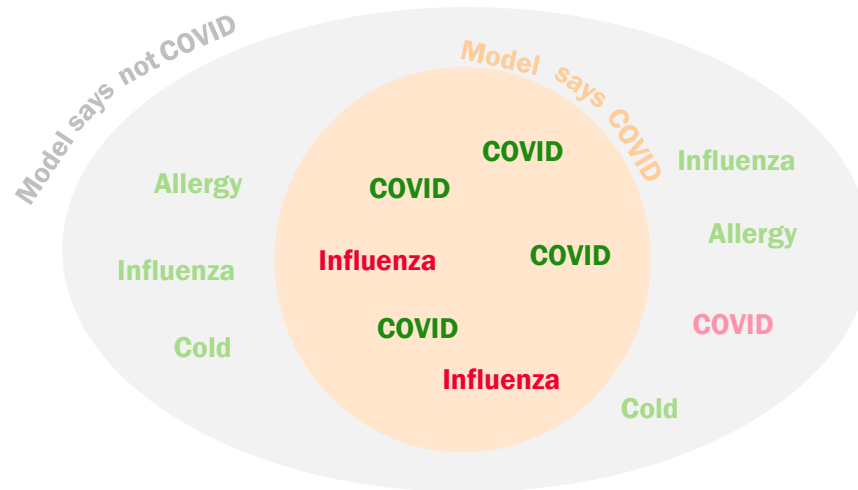
$$\text{Recall} = \frac{\text{relevant and retrieved}}{\text{relevant}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{■}}{\text{■} + \text{■}} = \frac{4}{4 + 1} = \frac{4}{5} = \mathbf{0.80}$$

$$F = \frac{2 \times P \times R}{P + R}$$

Evaluation

Classification

How well can we detect COVID?



- True positive
- True negative
- False positive
- False negative

$$\text{Accuracy} = \frac{\text{correct}}{\text{all}} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} = \frac{\text{True positive} + \text{True negative}}{\text{True positive} + \text{True negative} + \text{False positive} + \text{False negative}} = \frac{4 + 6}{13} = \frac{10}{13} = \mathbf{0.77}$$

$$\text{Precision} = \frac{\text{relevant and retrieved}}{\text{retrieved}} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{True positive}}{\text{True positive} + \text{False positive}} = \frac{4}{4 + 2} = \frac{4}{6} = \mathbf{0.67}$$

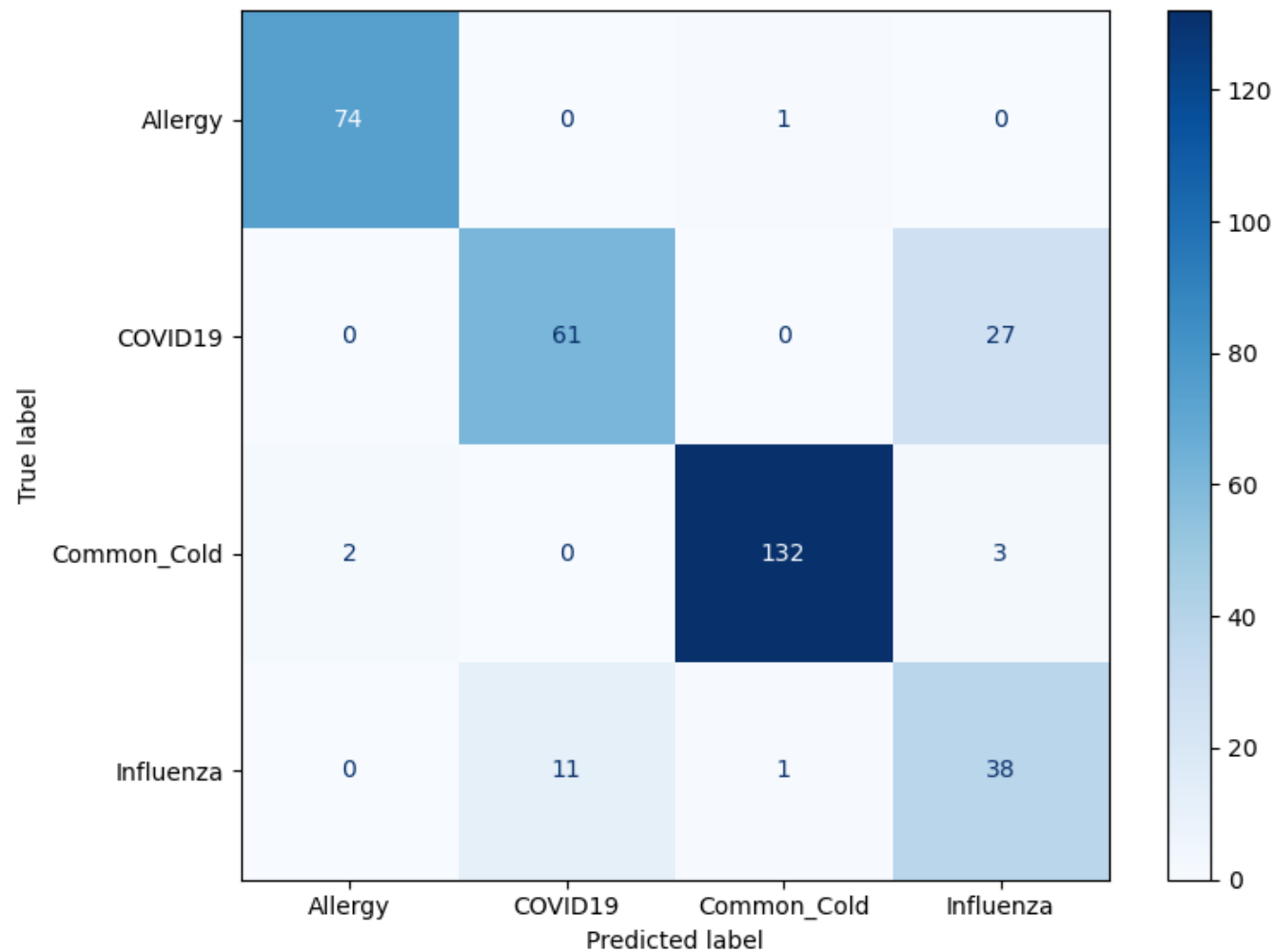
$$\text{Recall} = \frac{\text{relevant and retrieved}}{\text{relevant}} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{True positive}}{\text{True positive} + \text{False negative}} = \frac{4}{4 + 1} = \frac{4}{5} = \mathbf{0.80}$$

$$F = \frac{2 \times P \times R}{P + R} = \frac{2 \times 0.67 \times 0.80}{0.67 + 0.80} = \frac{1.072}{1.47} = \mathbf{0.73}$$

Evaluation

Classification

Confusion matrix



Evaluation Regression

E.g. How accurately can we predict how many days it will take a patient to recover from COVID?

Patient	Actual (y_i)	Predicted ($\hat{y}_i$)
1	10	12
2	14	13
3	7	8
4	21	15
5	18	17

Mean Average Error (MAE) $= \frac{1}{n} \sum |y_i - \hat{y}_i| = \frac{2 + 1 + 1 + 6 + 1}{5} = \frac{11}{5} = \mathbf{2.20 \text{ days}}$  **prediction error**

Root Mean Squared Error (RMSE) $= \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2} = \sqrt{\frac{4 + 1 + 1 + 36 + 1}{5}} = \sqrt{\frac{43}{5}} = \sqrt{8.60} \approx \mathbf{2.93 \text{ days}}$ 

Coefficient of Determination (R2) $= 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y}_i)^2} = \frac{43}{130} = 1 - 0.3308 = \mathbf{0.67}$

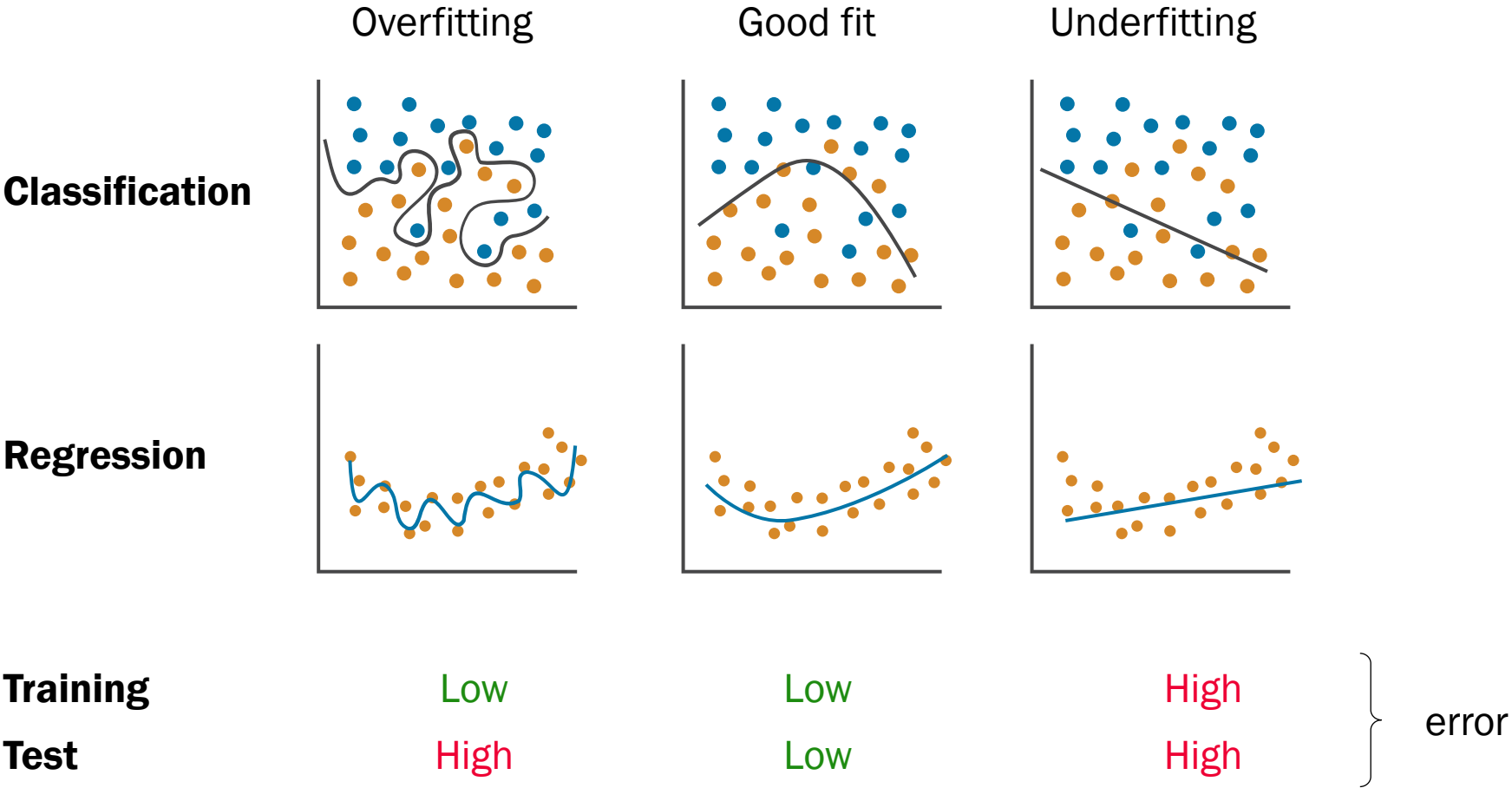
Evaluation

Specialist metrics

- **Automatic Speech Recognition (ASR) $\Rightarrow$ Word Error Rate (WER)**
How many word mistakes the speech-to-text model makes.
- **Machine Translation (MT) $\Rightarrow$ BLEU**
How closely a computer translation matches a human translation.
- **Text Summarisation $\Rightarrow$ ROUGE**
How much key information from the original text was kept in the summary.
- **Grammatical Error Correction (GEC) $\Rightarrow$ $F_{0.5}$ -Score**
Correction quality, heavily favoring precision over recall.
- **Search & Recommendation Engines $\Rightarrow$ Mean Reciprocal Rank / Normalized Discounted Cumulative Gain (NDCG)**
Whether the best answers appear at the top of the search list.



Overfitting and underfitting





See you on Friday!

Mariano.Felice@cl.cam.ac.uk